

ELECTROCHEMICAL CELLS.

(12)

An electrochemical cell consists of two electrodes, or metallic conductors, in contact with an electrolyte, an ionic conductor. An electrode and its electrolyte comprise an electrode compartment. The two electrodes may share the same compartment. The various kinds of electrodes are summarised below:-

Electrode type	Designation	Redox Couple	Half rxn.
1. Metal/metal-ion	$M(s)/M^{+}(aq)$	M^{+}/M	$M^{+}(aq) + e^{-} \rightarrow M(s)$
2. Gas electrode	$Pt(s)/X_2(g)/X^{+}(aq)$	X^{+}/X_2	$X^{+}(aq) + e^{-} \rightarrow \frac{1}{2} X_2(g)$
3. Metal/insoluble salt	$M(s)/MX(s)/X^{+}(aq)$	$MX/M, X^{-}$	$MX(s) + e^{-} \rightarrow M(s) + X^{-}$
4. Redox	$Pt(s)/M^{+}(aq), M^{2+}(aq)$	M^{2+}/M^{+}	$M^{2+}(aq) + e^{-} \rightarrow M^{+}(aq)$

Any inert metal shown as part of the specification is present to act as a source of sink of electrons (e^{-}) but takes no other part in the reaction other than acting as a catalyst for it. If the electrolytes are different, the two compartments may be joined by salt bridge, which is a tube containing a concentrated electrolyte solution in agar jelly that completes the electrical circuit and enables the cell to function. The electrolyte in the U-tube of salt bridge include such as NaCl, KCl or K_2SO_4 .

A galvanic cell is an electrochemical cell that produces electricity as a result of the spontaneous reaction occurring inside it. An electrolytic cell is an electrochemical cell in which a non-spontaneous reaction is driven by an external source of current.

How a Redox reaction can produce an electric current
When a bar of Zinc is dipped in a solution of copper sulphate, copper metal is deposited on the bar as shown below:

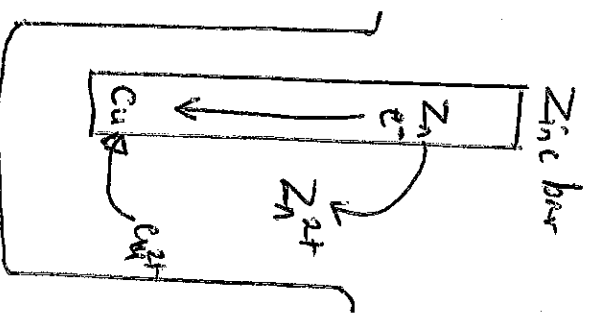


Fig. 1. Redox rxn taking place on zinc bar itself.

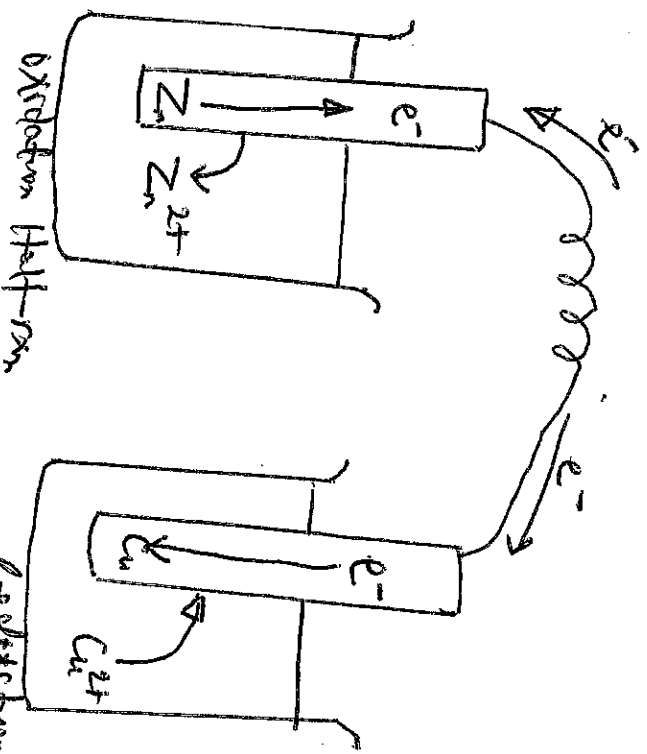
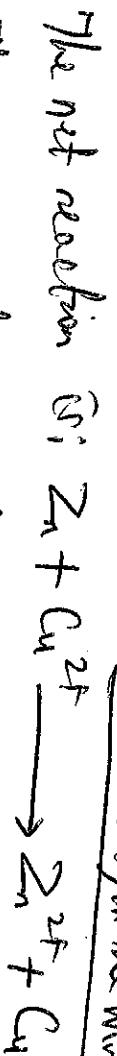


Fig. 2: Separate half-rxns cause flow of e^- (current) in the wire connecting the



This is a redox reaction and the two half-reactions are:



In this change, Zn is oxidized to give Zn^{2+} ions and Cu^{2+} ions are reduced to Cu atoms. The e^- released in

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The half-reaction are used up by the second half-reaction. Both the half-reactions occur on the Zinc bar itself and there is no net change.

Now, let the two half-reactions occur in separate compartments which are connected by a wire (Fig. 2). The e^- produced in the left compartment flow through the wire to the other compartment. Hence, the current will flow for an instant and then stop. The current stops flowing because of the charge build up in the two compartments. The e^- leave the left compartment and it would become truly charged. The right compartment receives e^- and become truly charged. Both these factors oppose the flow of e^- (current) which eventually stops.

The problem can be solved simply. The solution in the compartments may be connected by salt bridge - this provide a passage to ions from one compartment to the other without extensive mixing of the two solutions. With this, ion flow, the circuit is complete and e^- pass freely through the wire to keep the net charge zero in the two compartments.

Half-Reactions

Any redox reaction may be expressed as the difference of two reduction half-reactions, which are sometimes reactions showing the gain of e^- . For example, the reduction of Cu^{2+} ions by zinc can be expressed as the difference of the following two half-reactions.



The difference of the two (copper - zinc) is



Other reactions that are not redox reactions may often be expressed as the difference of two reduction half-reactions.

Exercise: Express the dissolution of silver chloride in water as the difference of two reduction half-reactions.

The reduced and oxidized species in a half-reaction form a redox couple. Thus the redox couple mentioned above for Cu^{2+}/Cu and Zn^{2+}/Zn . In general, we write a couple as OX/Red and the corresponding reduction half-reaction is

$$\text{OX} + \text{ve}^- \longrightarrow \text{Red}.$$

We shall often find it useful to express the composition of an electrode compartment in terms of the reaction

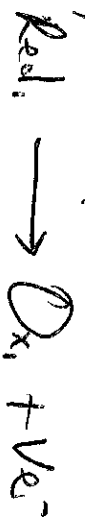
quotient, Q , for the half-reaction. This quotient is defined ⁽¹⁴⁾ like the reaction quotient for the overall reaction, but the e^- are ignored. Thus, for the copper half-reaction $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Cu}(s)$, we write:-

$$Q = \frac{1}{a_{\text{Cu}^{2+}}}.$$

We have used the fact that the pure metal (the standard state of the ~~element~~ element) has unit activity.

Reductions at Electrodes

The reduction and oxidation processes responsible for the overall reactions in a cell are separated in space: oxidation takes place in one electrode compartment and reduction takes place in the other compartment. As the reaction proceeds, the ~~ex~~ reduced in the oxidation at one electrode travel through the external circuit and re-enter the cell through the other electrode. Thus they bring about reduction.



The electrode at which oxidation occurs is called the anode; the electrode at which reduction occurs is called the cathode.

In a galvanic cell, the cathode has a higher potential than the anode: the species undergoing reduction, Ox_2 , with draws e^- from its electrode (the cathode, fig. 3), so leaving a relative $+$ ve charge on it (corresponding to a high potential). ~~(fig. 3)~~

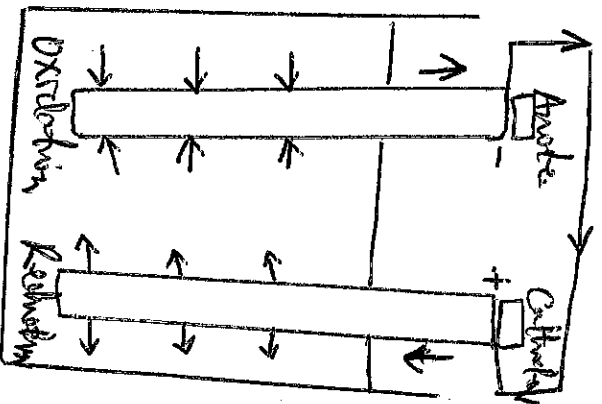


Fig. 3

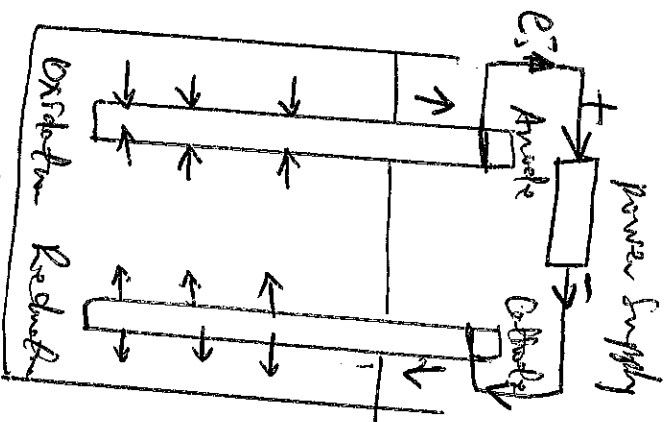


Fig. 4

At the anode, Oxidation results in the transfer of e^- to the electrode, so giving it a relatively $-$ ve charge (corresponding to a low potential). In an electrolytic cell, the anode is still the location of oxidation, but now e^- must be withdrawn from the species in that compartment because oxidation does not occur spontaneously, and at the cathode there must be a supply of e^- to drive the reduction. Therefore, in an electrolytic cell the anode must be made relatively $+$ ve to the cathode (fig. 4).

Liquid Junction Potentials

(15)

In a cell with two different electrolyte solutions in contact, as in the Daniell cell, there is an additional source of potential difference across the interface of the two electrolytes. This potential is called the liquid junction potential, E_j . Another example of a junction potential is that b/w different concentrations of hydrochloric acid. At the junction, the mobile H^+ ions diffuse into the more dilute solutions. The bulkier Cl^- ions follow, but initially do so more slowly, which results in a potential difference at the junction. The potential then settles down to a value such that, after that brief initial period, the ion diffuse at the same rate. Electrolyte combination cells always have a liquid junction; electrode concentration cells do not.

The contribution of the liquid junction to the potential can be reduced (to about 1-2 mV) by joining the electrolyte compartments through a salt bridge (Fig. 5)

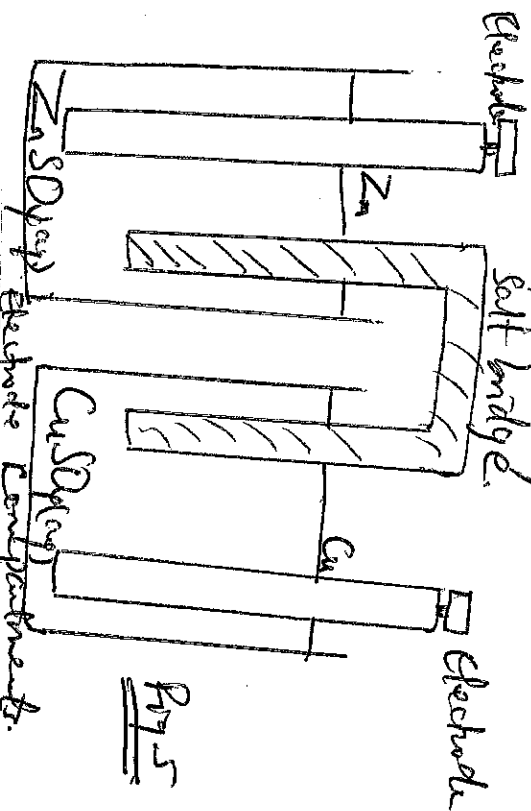
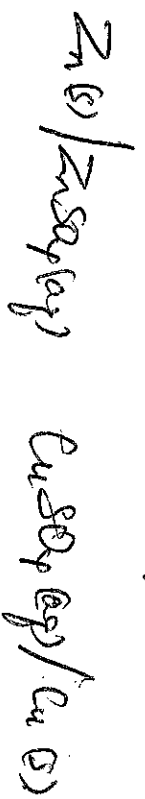


Fig. 5

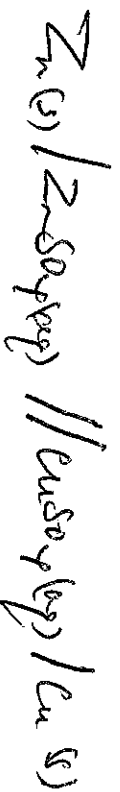
The reason for the success of the salt bridge is that the liquid junction potential at either end are largely independent of the conc. of the two dilute solutions and so nearly cancel.

Notation:

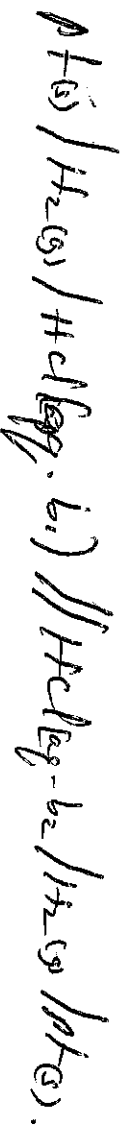
A liquid junction is denoted by:



A double vertical line $\parallel$ denotes an interface for which it is assumed that the junction potential has been eliminated.



An electrolyte concentration cell in which the liquid junction potential is assumed to be eliminated is denoted:

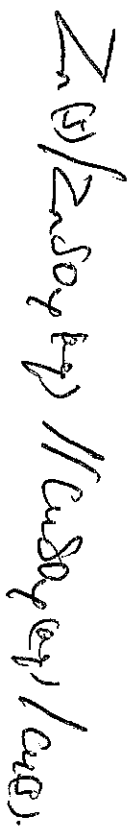


The Cell Reaction

The cell reaction is the reaction in the cell written on the assumption that the right-hand electrode is the cathode, and hence that the spontaneous reaction is one in which reduction is taking place in the right-hand compartment. If the left-hand electrode turns out to be cathode, then the reverse of the corresponding cell reaction is spontaneous.

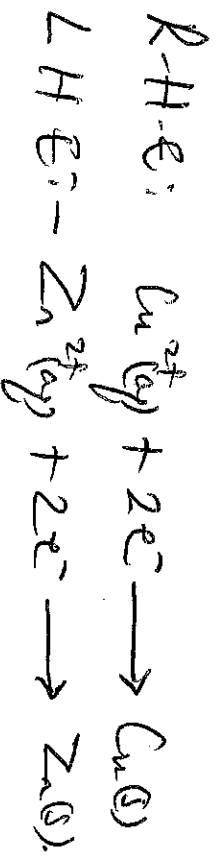
To write the cell reaction corresponding to a cell diagram, we first write the right-hand half-reaction

as a reduction (because we have assumed that to be spontaneous), then we subtract from it the left-hand reduction half-reaction (for, by implication, that electrode is the site of oxidation). Thus, in the cell,



the two electrode and their reduction half-reactions:

are:



Hence, the overall cell reaction is the difference:



• Cell Potential or Emf

A cell in which the overall cell reaction has not reached real equilibrium can do electrical work as the reaction drives e^- through an external circuit. Consider a Zn-Cu voltaic cell, e^- are released at the anode and it becomes very charged. The -ve electrode pushes e^- through the external circuit by electrical repulsion. The copper electrode gets +ve charge due to the discharge of Cu^{2+} ions on it. Thus e^- from the anode

Circuit are attracted into the electrodes the flow of current through the circuit is determined by the 'push' of E_s at the anode and 'attraction' of E_s at the cathode. These two forces constitute the 'driving force' or 'electromotive pressure' that sends e^- through the circuit. This driving force is called the electromotive force (emf) or cell potential. The cell potential is measured in Volts (V) and is also referred to as cell voltage.

When the cell potential is large, a given number of e^- travelling b/w the electrodes can do large ~~work~~ amount of electrical work and when small, the same number of electrons can do only a small amount of work. A cell in which the overall reaction is at equilibrium can do no work, and thus the emf is zero.

Sign Convention of emf

The magnitude of emf of a cell reflects the tendency of e^- to flow externally from one electrode to another. The e^- are transported through the cell solution by ions present and pass from the two electrodes to the ~~we~~ electrodes - this corresponds to a clockwise flow of e^- through the external circuit (as in Daniel cell). Thus, the emf of the cell is given the + sign. If the emf acts

in the opposite direction through the cell circuit, it is quoted as -ve value. For example, Daniel cell has an emf of 1.1V and the copper electrode is the.

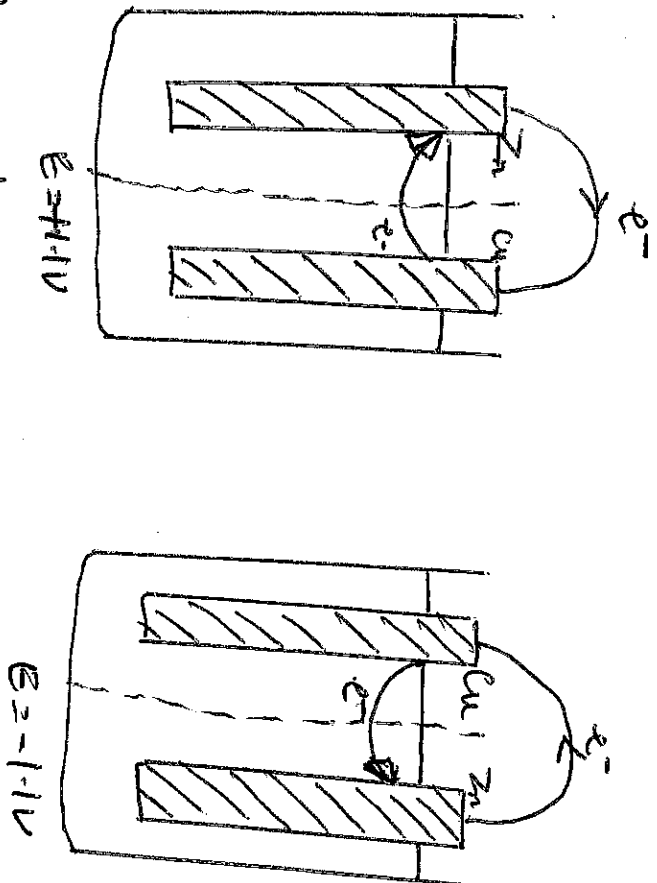
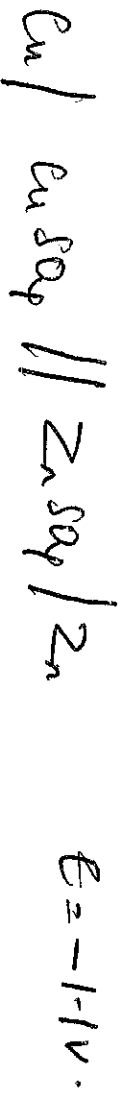
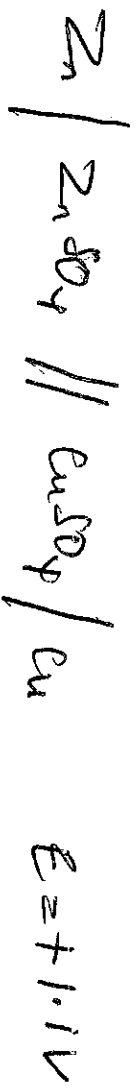


Fig. 61 Illustration of emf sign in Daniel cell

The above can be ~~it~~ expressed in two ways.



The -ve sign indicates that the cell is not feasible in the given directions. The reaction will take place in the reverse direction.

Calculating the emf of a cell

The emf of a cell can be calculated from the half-cell potentials of the two cells (anode and cathode) as follows

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \\ = E_R - E_L$$

where E_a and E_c are the reduction potentials of the LH and the LH electrodes respectively. It may be noted that absolute values of these reduction potentials cannot be determined. These are found by connecting the half-cell with a standard hydrogen electrode (SHE) whose reduction potentials have been arbitrarily fixed as 2.00.

How to Measure EMF of a cell

The emf of an unknown cell can be measured with the help of a potentiometer (Fig-7). It consists of a wire AB which is about a metre long. The two ends of this wire are connected to a working battery W.

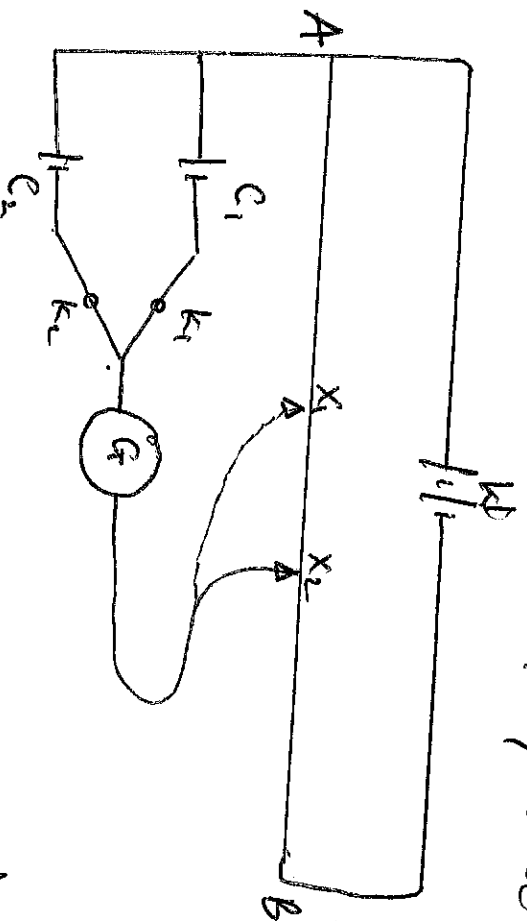


Fig. 7: Measuring the emf of a cell with a potentiometer

A standard cell E_1 (i.e. a cell of known emf) is connected to the end A. At the other end, the cell E_2 is connected to a galvanometer through a key K_1 . The galvanometer is then joined to a sliding contact that moves on the wire AB. The cell E_2 whose emf is to be measured is similarly connected to the key K_2 , the galvanometer and the sliding contact. By using the key K_1 , the cell E_1 is put into the circuit and the contact is moved to and fro along AB. When no current flows through the galvanometer, the point of contact X_1 is recorded. Then by using the key K_2 , the cell E_2 is put into ~~the~~ the circuit and the procedure is repeated to find the corresponding point X_2 . The emf of the cell E_2 is calculated by using the following equation:

$$\frac{\text{Emf of } E_2}{\text{Emf of } E_1} = \frac{\text{Distance } AX_2}{\text{Distance } AX_1}.$$

Cells emfs can also be measured by electronic Voltmeter of the analogue or digital type.

When the emf of a cell is determined under standard conditions, it is called the standard emf. The standard conditions are 1M solution of reactants and products and temperature of 25°C . It is represented by E° . With gases 1 atm pressure is a standard condition.

Reversible Cells.

A familiar example of a reversible cell is the Daniell cell (fig. 89). We know that e^- flows from Zinc electrode to Copper electrode due to the net cell reaction.

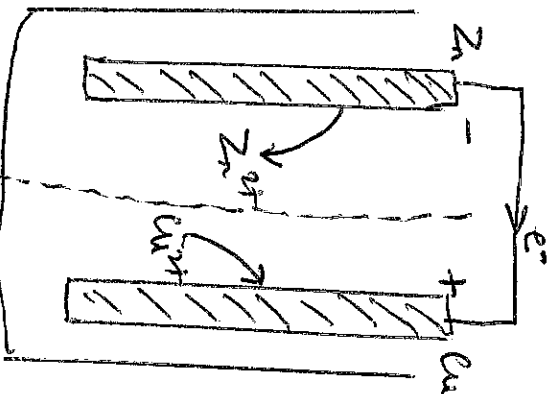


Fig. 89. Daniell's standard cell. Zn-Cu flows e^- from L to R

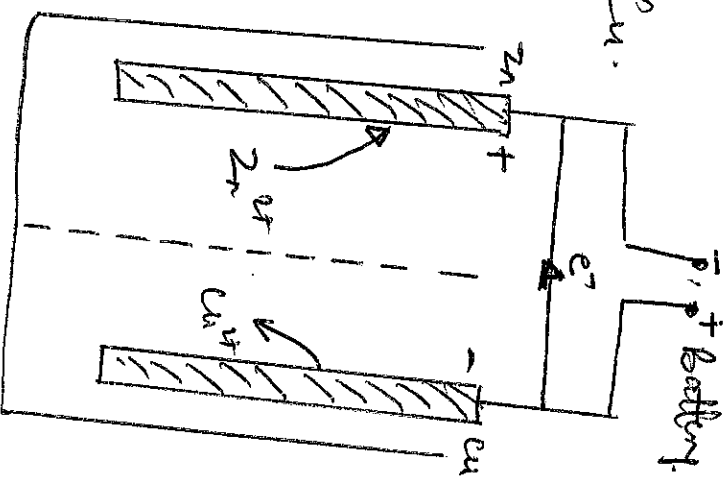
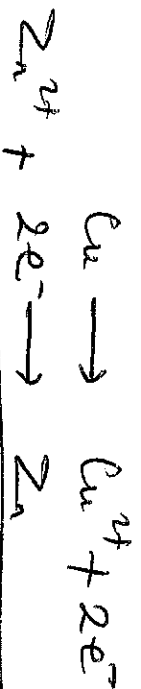


Fig. 86) Cu-Zn cell, current flows from R to L on application of external emf.

However, when the two electrodes are connected to an external battery that opposes the cell emf, the above reaction is reversed (fig. 86). Cu from the Copper electrode dissolves to form Cu^{2+} ion and Zn^{2+} ion is discharged on the Zinc electrode to give Zn atom. The overall reaction taking place in the cell may be written as



With the help of potentiometer (fig-7) the cell emf is exactly balanced by the external emf. At the balance point, no current flows through the circuit. Now let the external emf be decreased and then increased by an infinitesimal amount. A minute current will flow first to the right and then to the left. This reverse of the cell current is accompanied by a corresponding change in the direction of the cell reaction. This type of reversible behaviour is a feature of the reversible cells. A reversible cell may be defined as "A cell that operates by reversal of the cell current and direction of the cell reaction by infinitesimal change of emf on either side of the balance point."

Relation between emf and free energy.

When a cell produces a current, used to do work - to run a motor, for instance, thermodynamic principles can be employed to derive a relation b/w electrical energy and the maximum amount of work, W_{max} obtainable from the cell. The W_{max} from the cell is the product of charge flowing per mole and maximum potential difference, E , through which the charge is transported.

$$k_{\text{inner}} = -N_E \quad \text{---} \quad \text{(7)}$$

n is the no. of moles of e^- transferred and E is equal to the Voltage of the ion participating in the cell reaction. F is the Faraday = 96,500 coulombs and E is the EMF of the cell.

According to thermodynamics, the maximum Work that can be derived from a chemical reaction is equal to the free energy (ΔG) for the reaction.

$$W_{\max} = \Delta G \quad \text{--- (2)}$$

from (1) and (2) we can write:

$$\Delta G = -nFE \quad \text{--- (3)}$$

According to Gibbs-Helmholtz equation, the decrease in free energy of a system at constant pressure is given by the expression

$$-\Delta G = -\Delta H - T \left[\frac{\partial(\Delta G)}{\partial T} \right]_P \quad \text{--- (4)}$$

Substituting the value of ΔG from (3).

$$\begin{aligned} nFE &= -\Delta H - T \left[\frac{\partial(-nFE)}{\partial T} \right]_P \\ &= -\Delta H + nFT \left(\frac{\partial E}{\partial T} \right)_P \end{aligned}$$

$$E = \frac{-\Delta H}{nF} + T \left(\frac{\partial E}{\partial T} \right)_P.$$

THE NERNST EQUATION

We know experimentally that the potential of a single electrode or half-cell varies with the concentration of ions in the cell. In 1859 Walter Nernst derived a mathematical relationship which enables us to calculate the half-cell potential, E , from the standard electrode potential E° , and the temperature of the cell. This relation known as the Nernst equation can be stated as:-

$$E = E^\circ - \frac{2.303RT}{nF} \log K. \quad \text{--- (1)}$$

where E° = standard electrode potential

R = gas constant

T = kelvin temperature

n = no of e^- transferred in the half-reaction

F = Faraday of electricity

K = equilibrium constant for the half-reaction as in equilibrium

or $E = E^\circ - \frac{RT}{nF} \ln K$
 $\frac{1}{2} \text{ no of } e^-$
 $82K$

for an oxidation half-reaction when the metal electrode



The Nernst equation takes the form;

$$E = E^\circ - \frac{2.303RT}{nF} \log \frac{[M^{n+}]}{[M]} \quad \text{--- (2)}$$

The concentration of solid metal $[M]$ is zero. therefore,

Equation (2) becomes:-

$$E = E^\circ - \frac{2.303RT}{nF} \log [M^{n+}] \quad \text{--- (3)}$$

Substituting the values of R, F and T at 25°C, the quantity $2.303RT/F$ comes to be 0.0591. Hence, equation ③ can be written in its simplified form as:-

$$E = E^{\circ} - \frac{0.0591}{n} \log [M^{n+}] \quad \text{--- (4)}$$

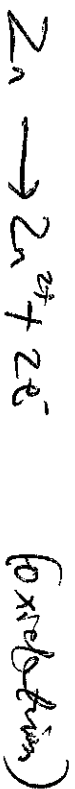
This is the equation for a half-cell in which oxidation occurs. In case it is a reduction reaction, the sign of E will be reversed.

Example,

What is the potential of a half-cell ~~containing~~^{consisting of} Zinc electrode in 0.01M $ZnSO_4$ solution at 25°C. $E^{\circ} = 0.763V$.

Solution,

The Half-cell reaction is,



The oxidation half-cell reaction from Nernst equation:-

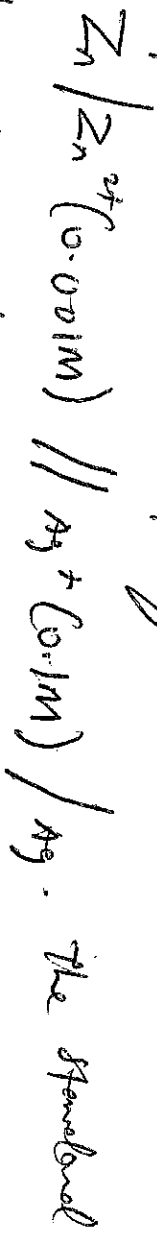
$$\begin{aligned} E &= E^{\circ} - \frac{0.0591}{n} \log [Zn^{2+}] \\ &= 0.763 - \frac{0.0591}{2} \log (0.01) \\ &= 0.763 + 0.0591 = \underline{\underline{0.8221V}} \end{aligned}$$

The Nernst equation is applicable to cell potential as well. Thus,

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log K \quad \text{--- (5)}$$

K is the equilibrium constant of the redox cell reaction.

Example 2: Cal. the E of the cell.



potential of Ag/Ag⁺ half-cell is +0.80V and Zn/Zn²⁺ is -0.76V.

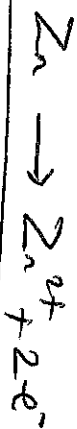
Solution:

The half-cell reaction of the anode and the cathode is

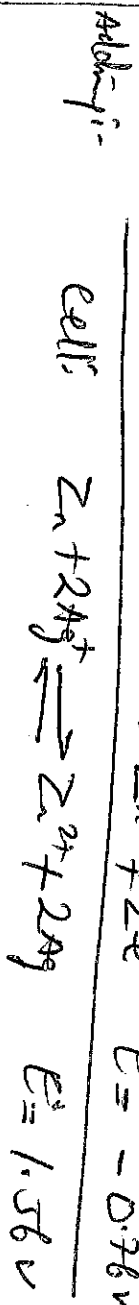


$$E^\circ = +0.80\text{V}$$

Anode:-



$$E^\circ = -0.76\text{V}$$



$$K \text{ for the cell reaction} = \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$$

Substitute the given values in the Nernst equation and deriving for E_{cell} , we have,

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log K$$

$$= 1.56 - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$$

$$= 1.56 - \frac{0.0591}{2} \log \frac{[10^{-3}]}{[10^{-1}]^2}$$

$$= 1.56 - 0.02955 (\log 10^{-1})$$

$$= 1.56 + 0.02955$$

$$= \underline{\underline{1.58955\text{V}}}$$

At equilibrium, the cell reaction is balanced and the cell potential is zero. The Nernst equation may, now, be written as:-

$$0 = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log K$$

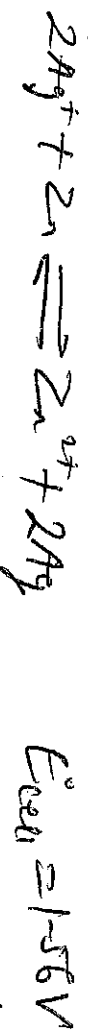
$$\text{or } \log K = \frac{n E_{\text{cell}}^{\circ}}{0.0591} //$$

Example 3:

Cal. the equilibrium constant for the reaction b/w silver nitrate and metallic zinc.

Solution:

The equation for the reaction:



Substituting values in the Nernst equation at equilibrium:-

$$\log K = \frac{n E_{\text{cell}}^{\circ}}{0.0591}$$

$$0 = 1.56 - 0.03 \log K$$

$$-1.56 = -0.03 \log K$$

$$\log K = \frac{-1.56}{-0.03} = 52$$

$$K = \underline{\underline{1 \times 10^{52}}}$$

Q1) 30 ml of $6.0 \times 10^{-3} M$ $CaCl_2$ is mixed with 30 ml of $0.04 M$ NH_4F . Will CaF_2 precipitate? K_{sp} of $CaCl_2 = 4.0 \times 10^{-11}$

Sol:

The rxn that can produce ppt is $Ca^{2+} + 2F^- \rightleftharpoons CaF_2(s)$

The conc of each ion in solution is:

$$[Ca^{2+}] = 6.0 \times 10^{-3} M \times \frac{30 \text{ ml}}{(30+30) \text{ ml}} = 3.0 \times 10^{-3} M$$

$$[F^-] = 0.040 M \times \frac{30 \text{ ml}}{60 \text{ ml}} = 0.015 M$$

$$Q = [Ca^{2+}][F^-]^2 = (3.0 \times 10^{-3})(0.015)^2 = \underline{\underline{6.75 \times 10^{-8}}}$$

Since $Q > K_{sp}$, ppt will occur.

Q2- Cal. the K_{sp} for bismuth sulphide (Bi_2S_3), which has a solubility of $1.0 \times 10^{-15} \text{ mol/L}$ at $25^\circ C$.

Sol The rxn is $Bi_2S_3 \rightleftharpoons 2Bi^{3+} + 3S^{2-}$

1 mole of Bi_2S_3 gives 2 mole of Bi^{3+} and 3 moles of S^{2-} ion. Thus 1.0×10^{-15} mole Bi_2S_3 will give $2(1.0 \times 10^{-15} \text{ mole})$ of Bi^{3+} and $3(1.0 \times 10^{-15} \text{ mole})$ of S^{2-} . We can write the rxn concs of Bi^{3+} and S^{2-} as:

$$[Bi^{3+}] = 2.0 \times 10^{-15} \text{ mol/L}$$

$$[S^{2-}] = 3.0 \times 10^{-15} \text{ mol/L}$$

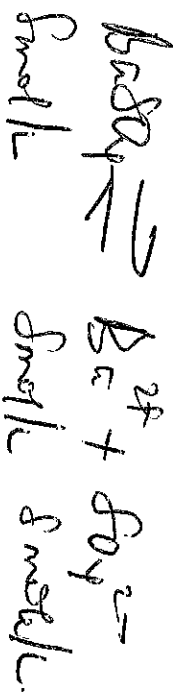
$$K_{sp} = [Bi^{3+}]^2 [S^{2-}]^3$$

$$= (2.0 \times 10^{-15})^2 (3.0 \times 10^{-15})^3$$

$$= \underline{\underline{1.08 \times 10^{-73}}}$$

Q3. The solubility of $CaSO_4$ is $2.3 \times 10^{-3} \text{ g/ml}$ at $20^\circ C$. Cal. the solubility product of $CaSO_4$, assuming that the salt is completely ionized.

Sol



Value of K_{sp} the solubility of BaSO_4 at 20°C .

$\therefore$ Amount of BaSO_4 in one ml $\geq 2.3 \times 10^{-5} \text{ g}$.

$$\checkmark \quad \checkmark \quad \text{one L} = 2.3 \times 10^{-5} \times 10^3 \text{ g}$$

$$\therefore \text{no. of moles of } \text{BaSO}_4 \text{ per litre} = \frac{2.3 \times 10^{-5} \times 10^3}{233} = 1 \times 10^{-6}$$

$$\therefore K_{sp} = [\text{Ba}^{2+}] [\text{SO}_4^{2-}]$$

$$= (1 \times 10^{-6}) (1 \times 10^{-6})$$

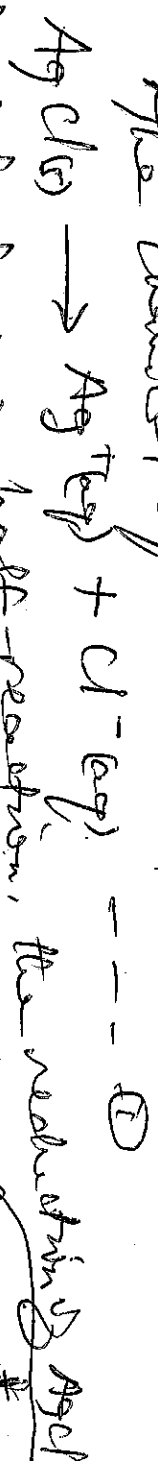
Thus the solubility product of BaSO_4 at 20°C is 1×10^{-12}

Solution to an Exercise:

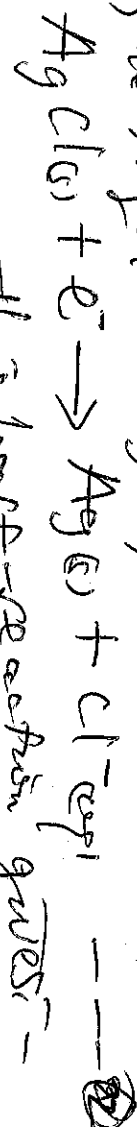
Express the direction of silver chloride in water as the difference of two reduction half-reactions

Solution:

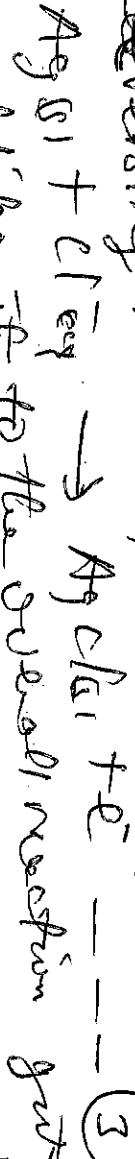
The standard reduction of the overall reaction is



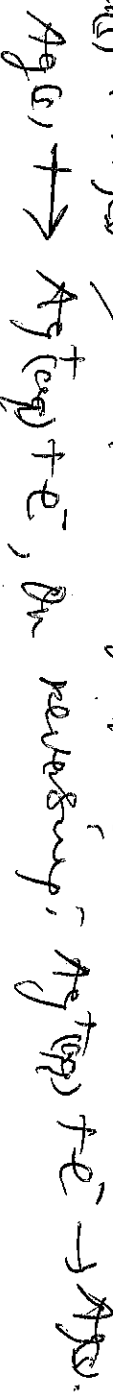
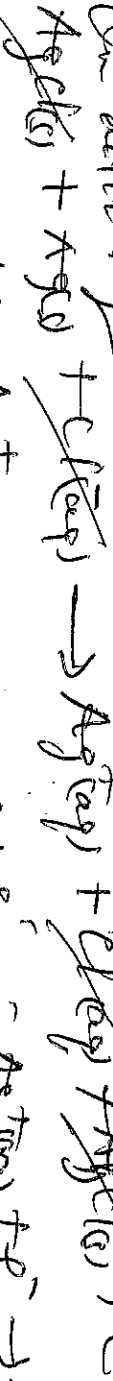
We select one half-reaction, the reduction of AgCl to Ag and Cl^- .



Reversing this half-reaction gives



On adding it to the overall reaction gives



Comment: There is no net change of mass in the reaction, so no net change of mass.